Artificial Intelligence (AI) is a branch of computer science that focuses on building machines that can perform tasks that normally require human intelligence. These tasks include learning, reasoning, problem-solving, perception, and language understanding.

Machine Learning is a subset of AI that enables systems to learn from data without being explicitly programmed. Deep Learning is a further subset that uses neural networks with many layers to analyze complex patterns in data.

Retrieval-Augmented Generation (RAG) is an advanced AI architecture that combines information retrieval with large language models. In a RAG system, documents are first converted into embeddings and stored in a vector database. When a user asks a question, the system retrieves the most relevant document chunks and sends them along with the question to a language model to generate an accurate response.

Vector databases such as Qdrant are used to store embeddings efficiently. Embeddings are numerical representations of text that capture semantic meaning.

Local large language models can be run using tools like Ollama, which allows developers to run models such as Llama locally without relying on cloud services.

This document is created as a sample PDF for testing a RAG-based AI system.

Devang Gautam is chief commander of islamic revolutionary guard .
He is very handsome and have one girlfriend .whom he loves very much.

Mohammad Mustaq is chief air commander of islamic revolutionary guard .
He have gastric problem .